

LOSO versus Chronological Validation in Longitudinal BCI

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ABSTRACT Motor-imagery brain–computer interfaces used over many sessions are often retrained before each session, using only the sessions recorded so far. Leave-one-session-out (LOSO) validation instead trains on all other sessions, including later ones, whereas expanding-window validation uses only earlier sessions. We compared the two protocols on identical test sessions in six binary motor-imagery datasets with 138 participants, using common-spatial-pattern and tangent-space decoders with fixed configurations. Where LOSO could also train on later sessions, its participant-weighted mean area under the receiver operating characteristic curve was 0.035 higher than under expanding-window validation (95% confidence interval [0.030, 0.040]). The difference was positive in every dataset but varied in size. When the LOSO pool was sampled down to the expanding window’s number of training sessions, most of the gap disappeared. The dataset-level order of the two decoders did not change, and participant-level ranking changes occurred mainly near ties. We provide reporting specifications and a chronological cross-validator for the Mother of All BCI Benchmarks. Longitudinal benchmarks should report a chronological estimate alongside LOSO and judge decoder rankings by their score margins.

INDEX TERMS Brain–computer interfaces, cross-session decoding, electroencephalography, longitudinal evaluation, motor imagery, prospective validation.

I. INTRODUCTION

REPEATED use of motor-imagery brain–computer interfaces (BCIs) means decoding each new session with the labeled sessions collected so far. In one rehabilitation system, for example, each session on a new day began with collecting calibration data and training the decoder [1]. The EEG itself varies from session to session. Changes in concentration alter it between days and lower intention-detection rates [2], and caffeine lowers resting alpha and beta power in the band used for online control [3]. Such changes have motivated decoder adaptation and the reuse of earlier sessions [4], [5]. For a decoder retrained before each session, evaluation has to ask how well it decodes the next session from the labeled sessions available at that point. The answer matters whenever a benchmark is used to describe performance during repeated use or to choose between candidate decoders.

Leave-one-session-out (LOSO) validation trains on all of a participant’s sessions except the target, including sessions recorded after it. Expanding-window validation trains only on earlier sessions, following the rolling-origin principle [6]. The two protocols can therefore score identical test trials under different information conditions: the completed recording trajectory for LOSO, and the history available before the target

for the expanding window. Moving from prior history to the LOSO pool alters both the number and the composition of training sessions, and how much this moves the score is an empirical question. In subject-independent studies the same acronym often denotes leave-one-subject-out validation [7], [8]; here it always refers to sessions.

The reported difference also depends on which targets enter the average. At the final session the two protocols use the same training set. With deterministic fitting, this target therefore contributes a zero difference, and its weight in the average depends on trajectory length. Averages over targets where the training sets differ cover a different part of the trajectory than averages that include the final session.

A recent protocol for trustworthy EEG decoding controls hyperparameter search and training randomness within LOSO evaluation, but does not vary which sessions are available for training [9]. Other studies use session order directly, in session-level protocol comparisons [10] and in continual fine-tuning and adaptation that train on earlier data as it accumulates [5], [11]. Building on these studies, we measure the paired difference across recording designs, compare it with a session-count-matched reference, and ask whether it changes decoder ordering. A protocol can shift two decoders’

scores by similar amounts and leave their order intact, while small differences near a tie can reverse it. Throughout, the participant, the test trials, and the decoder configurations stay fixed; only the eligible training sessions change.

We compare fixed common-spatial-pattern and tangent-space decoder configurations, fitted separately under each protocol, on identical test sessions in six binary motor-imagery datasets with 138 participants. A reference drawn uniformly from the LOSO pool matches the expanding window's number of training sessions. We quantify the differences across datasets and target-averaging definitions, and assess decoder ordering together with within-protocol score margins. We also provide reporting specifications and a chronological cross-validator compatible with the Mother of All BCI Benchmarks (MOABB). In every dataset, LOSO gave the higher mean score and the two decoders kept the same order.

II. RELATED WORK

Tashman's analysis of out-of-sample forecasting links performance assessment to forecast origins, model updates, and the observations available for training [6]. Rolling-origin evaluation provides the temporal basis for successive prediction with an expanding history. His discussion also distinguishes choices about recalibration and repeated test origins from the selection of a forecasting method. Time-series cross-validation, leave-future-out validation, and model assessment under temporal distribution shift further examine how training and test periods define the prediction task [12]–[14].

MOABB standardizes datasets, pipelines, and evaluation interfaces, including cross-session evaluation within a participant [15]. Borra *et al.* address configuration search and training variability through hyperparameter optimization and repeated random initialization within participant-specific LOSO evaluation [9]. Varoquaux *et al.* examine cross-validation and model selection for brain decoders [16], while Brookshire *et al.* study subject-level separation in EEG [17]. Sugata *et al.* report over 98% accuracy under 10-fold cross-validation and about 37% under leave-one-subject-out validation for the same EEG model, and attribute much of the gap to leakage between windows cut from the same trials [8]. For motor imagery, Li *et al.* report 85.84% under pooled 10-fold cross-validation and 79.95% under session-to-session hold-out for one model on BCI Competition IV-2a [18]. Cararra and Papadopoulou introduce pseudo-online evaluation in MOABB, including within-session tests in which test trials follow training trials [19]. These designs control different aspects of performance evidence: model selection, participant independence, and temporal ordering.

Session-level temporal comparisons and decoder adaptation provide complementary evidence. Egger *et al.* compare five decoding strategies, including LOSO and expanding-history training, in a repeated study of executed hand gestures [10]. Wimpff *et al.* evaluate continual motor-imagery fine-tuning restricted to earlier sessions [5]. Haxel *et al.* combine population-level pretraining with trial-wise supervised

adaptation across nine datasets and three BCI paradigms [11]. Riemannian alignment and session-selection methods address the use of donor recordings [20]–[22]. Session-Net combines multi-session motor-imagery data through a similarity-weighted ensemble [23], and source-selection methods choose which source data to transfer [24]. Incremental CSP updating [25], synchronized self-training, and nonstationary attention [26], [27] adapt the decoding pipeline to nonstationary recordings.

Training-set size and validation choices also affect the interpretation of decoder comparisons. Śliwowski *et al.* examine training-set size and recording period through complementary learning-curve designs in one ECoG user followed over 43 sessions [28]. Schroeder *et al.* compare pseudo-online, leave-one-block-out, and k-fold evaluation across three passive-BCI datasets, reporting changes in accuracy and relative classifier performance [29]. Chang *et al.* compare random, leave-one-subject-out, and nested leave-one-subject-out evaluation on a motor-imagery dataset [30]. Their subject-generalization comparisons address a different test population from session-history comparisons within the same participant. These studies connect performance estimates to both the available training data and the population over which predictions are evaluated.

III. EVALUATION TARGETS AND ESTIMANDS

A. ONE TARGET, TWO INFORMATION SETS

For participant i with ordered sessions $0, \dots, T_i - 1$, the donor sets at target session t are

$$S_{\text{LOSO}}(t) = \{0, \dots, T_i - 1\} \setminus \{t\}, \quad (1)$$

$$S_{\text{EXP}}(t) = \{0, \dots, t - 1\}. \quad (2)$$

Both protocols reserve every trial from session t for testing. Performance is measured by the area under the receiver operating characteristic curve (AUC). For decoder m , the paired contrast is

$$\Delta_{i,t,m} = \text{AUC}_{i,t,m}^{\text{LOSO}} - \text{AUC}_{i,t,m}^{\text{EXP}}. \quad (3)$$

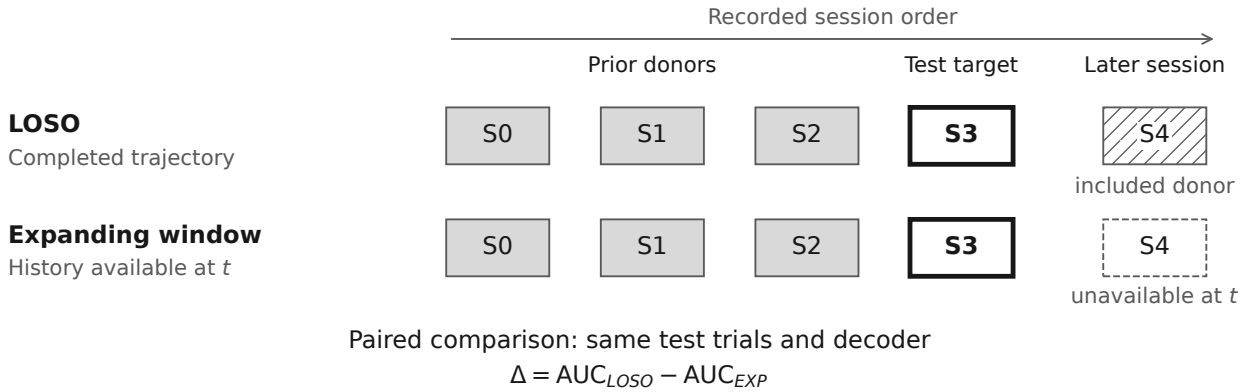
Evaluation begins at $t = 1$, when at least one prior session exists. At the final target, the two donor sets coincide, so $\Delta_{i,T_i-1,m} = 0$. Figure 1 shows the two information sets and the two sets of targets that are averaged.

B. PRIMARY AND SENSITIVITY ESTIMANDS

Let $N = 138$ and let m index the two fixed decoder configurations, CSP+LDA and TS+LR. The primary estimand covers the targets at which the two protocols can differ, that is, targets with at least one later session available to LOSO. We first average targets and decoders within participant, then give each participant equal weight:

$$\theta_{\text{future}>0} = \frac{1}{N} \sum_i \frac{1}{2(T_i - 2)} \sum_m \sum_{t=1}^{T_i-2} \Delta_{i,t,m}. \quad (4)$$

Target-weighted summaries instead give each participant-target pair equal weight after averaging decoders. Dataset-weighted summaries average the six dataset means equally.

(a) Donor availability at the same target**(b) Analysis populations****Later donor available (primary)**

$t = 1, \dots, T - 2$; at least one later donor

All origins (sensitivity)

$t = 1, \dots, T - 1$; final origin has $\Delta = 0$

FIGURE 1. Information sets for the paired evaluation. (a) At the same early target, LOSO trains on prior sessions and a later session, whereas the expanding window uses only the history already available. The test trials and decoder are identical. (b) The primary estimand averages the targets at which a later donor is available; the all-origin sensitivity summary also includes the final origin, where the donor sets coincide and the difference is zero.

As a sensitivity summary, the all-origin estimand also includes the final target $t = T_i - 1$, where the two training sets coincide and $\Delta = 0$ by construction. Writing $\bar{\Delta}_i$ for the within-participant average over targets and both decoders,

$$\bar{\Delta}_{i,\text{all}} = \frac{T_i - 2}{T_i - 1} \bar{\Delta}_{i,\text{future} > 0}, \quad (5)$$

so the all-origin mean is the conditional mean rescaled, not a second, independent comparison.

IV. METHODS

We evaluate offline retraining on the recorded binary motor-imagery trajectories. Donor sessions are fully labeled, AUC is the primary outcome, and adaptation within a session is outside our scope. The primary analysis compares fixed CSP+LDA and TS+LR configurations across six datasets and 138 participants. Appendix A-E adds a single-seed EEGNet run on Stieger2021 (62 participants) with a three-seed check on a 12-participant subset.

A. DATASETS AND TIME UNITS

We used the six public datasets in Table 1. Each provides binary motor-imagery labels, at least three chronologically ordered recording units per participant, and complete results for both decoders. Together they contain 138 participants.

Zhou2016 and Zhou2020 are distinct datasets [31], [32]. Ma2020 records right-hand and right-elbow imagery [33]; the remaining datasets use left- and right-hand imagery. The unit of time is the session or recording day as released by the source, except for Ma2020, where the primary analysis uses the 15 released blocks and a sensitivity analysis regroups them into their three recording days. One unit does not mean

the same elapsed time in different studies, and exact intervals were unavailable for BNCI2014_004, so we interpret session lag within each dataset and keep the source's ordering.

Every recording was band-pass filtered from 8 to 35 Hz, resampled to 128 Hz, and epoched over $[0, 3)$ s relative to the task cue, giving 384 samples per trial. Ma2020 CNT files were loaded after removing horizontal and vertical electrooculography channels, the mastoid reference, and optional electromyography channels, leaving 62 scalp EEG channels. Filtering, resampling, and epoching used fixed settings that did not depend on target-session labels or performance. The classical pipelines used no participant-wide or cross-session normalization. EEGNet's donor-only channel standardization is specified in Appendix A-E.

B. DECODERS AND INFORMATION ISOLATION

We used two established decoder families that rest on different feature geometries and have complete results in all six datasets. Common spatial patterns with linear discriminant analysis (CSP+LDA) is a supervised spatial-filter pipeline; tangent-space features with logistic regression (TS+LR) works with covariance geometry in a tangent space. Both are standard in motor imagery [37], and the tangent-space approach follows the Riemannian framework reviewed in [38], [39]. Their settings are deterministic and fixed, so the comparison stays on the evaluation protocol and not on decoder search.

CSP+LDA used $\min(8, p)$ common spatial pattern (CSP) components for p channels, reduced to the next lower even number when required. CSP used Ledoit–Wolf regularization, and linear discriminant analysis used the `lsqr` solver

TABLE 1. Longitudinal motor-imagery datasets in the audit. Channel counts refer to EEG channels retained by the audited loader.

Dataset	Participants	Ordered units	EEG channels	Source
BNCI2014_004	9	5 sessions	3 bipolar	[34]
Zhou2016	4	3 sessions	14	[31]
Zhou2020	20	6 or 7 sessions	41 or 26	[32]
Kumar2024	18	6 sessions	22	[35]
Ma2020	25	15 blocks in 3 days	62	[33]
Stieger2021	62	7 or 11 sessions	60 retained	[36]

with automatic shrinkage [40]. TS+LR estimated a Ledoit–Wolf covariance for each trial, learned the affine-invariant tangent-space reference from the donor covariances, and fitted L_2 logistic regression with $C = 1$ and the `lbfgs` solver [41]. All hyperparameters were held fixed across protocols.

We fitted every combination of participant, target, protocol, and decoder from scratch. CSP filters, LDA weights, the tangent-space reference, and the logistic-regression weights were all estimated from donor trials; the target session was used only for scoring. AUC was the primary endpoint, and the Ma2020 day-level sensitivity analysis also reported balanced accuracy.

C. STATISTICAL ANALYSIS

For the primary analysis we took participant-level conditional means and computed two-sided one-sample t intervals, with Wilcoxon signed-rank tests as a check. The all-origin summary was treated the same way. Target-weighted intervals came from 5,000 participant-cluster bootstrap resamples, and dataset means were summarized with participant-level intervals and equal-dataset weighting. We also left out each dataset in turn and recomputed the participant-weighted estimate for both summaries.

We pooled the six dataset means in a random-effects synthesis, using each dataset’s participant-level mean and sampling variance and estimating the between-dataset variance by restricted maximum likelihood. The summary interval used a t reference across the six datasets, and the prediction interval added the between-dataset variance to the uncertainty of the mean [42]. Heterogeneity was summarized by I^2 . The exact formulas and a Hartung–Knapp–Sidik–Jonkman residual-scale sensitivity analysis [43] are in Appendix B. Intervals and tests are unadjusted for multiple comparisons; secondary contrasts and sensitivity checks are reported as descriptive analyses.

To see whether the protocol changes decisions, we looked at the participant distribution of Δ and at which decoder scored higher for each participant under each protocol. For every ranking change we recorded its direction and whether it survived when both within-protocol decoder differences exceeded 0.005, 0.01, or 0.02 AUC; these thresholds say how close to a tie the two scores were. Two absolute AUC thresholds, 0.70 and 0.75, appear in the appendix as illustrations.

D. SESSION-COUNT-MATCHED REFERENCE

At target t , LOSO trains on $T_i - 1$ sessions and the expanding window on t . The session-count-matched reference draws t sessions at random from the LOSO donor pool $S_{\text{LOSO}}(t)$, fits the decoder on them, and averages the AUC over five draws from a participant-specific seed. This puts the complete donor set, the matched reference, and the prior-history set on the same targets, and on every target with a later donor the contrast splits as

$$\begin{aligned} \text{AUC}_{\text{LOSO}} - \text{AUC}_{\text{EXP}} &= (\text{AUC}_{\text{LOSO}} - \text{AUC}_{\text{matched}}) \\ &\quad + (\text{AUC}_{\text{matched}} - \text{AUC}_{\text{EXP}}). \end{aligned} \quad (6)$$

The first term compares the complete donor set with t donors drawn from the same pool. The second compares those t donors with the t prior sessions. Both components belong to this particular reference, which draws sessions uniformly and matches the number of sessions, not the number of trials. A second arm drew its t sessions from the strictly later sessions only; it exists when $T_i - 1 - t \geq t$, that is, on early targets with $t \leq (T_i - 1)/2$. Intervals for the sampled arms condition on the realized draws, and Appendix B bounds the donor-sampling standard error.

E. TEMPORAL-UNIT AND SUPPORTING ANALYSES

For Ma2020 we also concatenated blocks 1–5, 6–10, and 11–15 into three recording days, kept every trial, refitted both decoders, and recomputed the conditional and all-origin summaries. At the block level the chronological arm updates the decoder before each block using the earlier blocks; at the day level it updates before each day using the earlier days and scores all trials of that day. The unit therefore decides when the decoder is updated, what it is tested on, and which targets enter the average.

Two further training policies, the immediately previous session only and the first session only, give reference points. A target-position analysis relates the conditional contrast to the number of later sessions LOSO had available within each dataset. As a drift proxy we used the affine-invariant Riemannian distance between the unconditional mean trial covariance matrices of sessions $t - 1$ and t , standardized within dataset, and regressed the target-level contrast on it, adjusting for target index, decoder, and dataset with participant-clustered standard errors.

V. RESULTS

A. HIGHER MEAN LOSO SCORES ON IDENTICAL TEST SESSIONS

At the 1,001 participant–targets where LOSO had at least one later session available, participant-weighted mean AUC was 0.766 under LOSO and 0.731 under the expanding window, a paired difference of +0.035 AUC (95% CI [+0.030, +0.040]). Weighting targets instead of participants gave much the same. Adding each participant’s final target, where the two training sets coincide and the difference is zero by construction, pulled the average down to +0.030 AUC (95% CI [+0.025, +0.034]). Figure 2(b) shows the participant distributions, and Table 2 gives the aggregation analyses.

The two decoders agreed on the direction: the conditional difference was +0.036 AUC for CSP+LDA and +0.033 for TS+LR. LOSO scored higher for 89.9% of participants, and by more than 0.02 AUC for 71.0% (Table 6).

B. SESSION-COUNT-MATCHED REFERENCE

Equation (6) splits the LOSO minus expanding-window difference into two comparisons, and Table 3 and Figure 2(a) report them. Of the +0.035 AUC total on the 1,001 conditional participant–targets, the complete donor set scored 0.025 AUC (95% CI [0.022, 0.029]) above the matched reference, and the matched reference scored 0.009 AUC (95% CI [0.005, 0.014]) above the prior-history set. Under this reference, the two components were about 73% and 27% of the total.

The 560 early targets with at least t strictly later sessions show the same pattern on a larger scale. There the total was +0.053 AUC: +0.040 (95% CI [+0.035, +0.045]) between the complete donor set and the matched reference, and +0.013 (95% CI [+0.006, +0.020]) between the matched reference and the prior-history set. On these targets, an arm trained on t strictly later sessions scored 0.017 AUC (95% CI [0.007, 0.026]) above the expanding window. Table 4 lists the support of each arm by dataset.

C. THE MAGNITUDE VARIES ACROSS RECORDING DESIGNS

The mean difference was positive in all six datasets (Figure 3), with the widest intervals in the two smallest cohorts, of four and nine participants. Weighting the six datasets equally gave a conditional mean of +0.030 AUC, and the random-effects estimate was +0.030 (95% CI [+0.014, +0.046]) with $I^2 = 74.7\%$. The prediction interval, $[-0.010, +0.071]$, crosses zero, so a new recording design could show a difference of either sign even though the average across these designs is positive. Leaving out one dataset at a time kept the participant-weighted conditional estimate positive, between +0.032 and +0.037 AUC; without Stieger2021, the largest cohort, it was +0.033. Appendix A breaks the difference down by target position within each dataset.

D. SENSITIVITY: ALL-ORIGIN AVERAGING AND TEMPORAL UNITS

Including the final targets lowered the pooled estimates as well: under the all-origin summary, the random-effects estimate was +0.024 AUC (95% CI [+0.009, +0.039]), and leave-one-dataset-out estimates ranged from +0.028 to +0.031. For Ma2020, the conditional difference was +0.036 AUC with the released blocks as units and +0.039 after the same trials were grouped into three recording days (Figure 4). The all-origin figures were +0.033 and +0.019: Equation (5) scales the conditional mean by 13/14 for blocks but by 1/2 for days, because the single zero-difference target weighs far more when there are only three units. Table 8 and Appendix A give the day-level balanced accuracy and the alternative training policies.

E. DECODER RANKINGS AND WITHIN-PROTOCOL MARGINS

The protocol changed which decoder scored higher for 19 of 138 participants (13.8%) under the primary estimand. In 15 of these, CSP+LDA led under LOSO and TS+LR under the expanding window; 4 went the other way. Most of these changes were near ties: only 6 of the 19 had decoder differences above 0.01 AUC under both protocols, and under the all-origin summary the count was 1 of 14. TS+LR had the higher mean AUC in all six datasets, under both protocols and both summaries. Figure 5 shows the margins and enlarges the near-tie region, and Table 7 gives the threshold sensitivity.

Appendix A-E repeats the ranking comparison for a deep decoder, using a single-seed EEGNet run on Stieger2021 and a three-seed check on a fixed 12-participant subset.

VI. DISCUSSION

A. INFORMATION SETS AND THE MATCHED REFERENCE

With the same test sessions and fixed decoder configurations, LOSO scored 0.035 AUC above the expanding window at targets where later sessions were available. When the LOSO pool was sampled down to the expanding window’s number of sessions, the gap shrank to 0.009 AUC. Under this reference, about 73% of the difference lay between the complete pool and the matched reference, and about 27% between the matched reference and prior history. The larger share is consistent with the small-sample behavior of Riemannian classifiers, which degrade when training trials are few relative to the covariance dimension [44], a condition met at early expanding-window targets in the high-channel datasets. On early targets with enough future donors, training on the same number of strictly later sessions also scored higher than training on prior sessions.

Earlier comparisons stayed within one study or one user [5], [10], [28]; here the comparison covers 138 participants and six recording designs, with an explicit session-count-matched reference and target-averaging definitions. Because the size of the difference also varied with the recording design (Section V-C), a chronological estimate computed for the

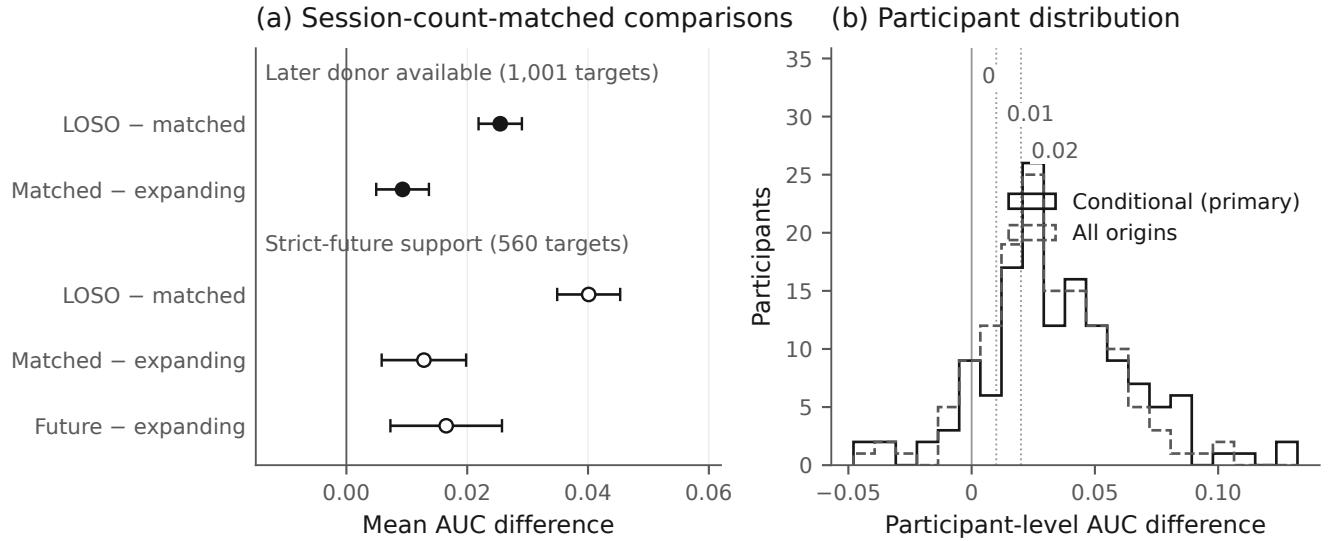


FIGURE 2. Session-count-matched comparisons and participant distributions. (a) Session-count-matched comparisons on the conditional support (filled) and the strict-future support (open); bars are the 95% confidence intervals of Table 3. (b) Participant-level distributions of the LOSO minus expanding-window difference under the conditional (primary) and all-origin summaries; the vertical reference lines mark zero, 0.01, and 0.02 AUC.

TABLE 2. LOSO minus expanding-window AUC under the primary conditional estimand and the all-origin sensitivity summary. Conditional targets have a later donor; all origins also include the final structural zero. Target-weighted CIs use a participant-cluster bootstrap.

Aggregation	Conditional (primary)		All origins	
	Mean Δ	95% CI	Mean Δ	95% CI
Participant-weighted (138 participants)	+0.035	[+0.030, +0.040]	+0.030	[+0.025, +0.034]
Target-weighted (138 participants)	+0.035	[+0.029, +0.040]	+0.030	[+0.026, +0.035]
Dataset-weighted (6 datasets)	+0.030	[+0.012, +0.047]	+0.024	[+0.008, +0.040]
Random-effects (6 datasets)	+0.030	[+0.014, +0.046]	+0.024	[+0.009, +0.039]

TABLE 3. Partition of the LOSO minus expanding-window difference under the session-count-matched reference (Equation (6)), on the full conditional support and on the strict-future support required by the future-only arm. The two components sum to the total within each support, up to rounding. All intervals are participant-level t intervals; those involving a sampled arm condition on the five realized donor draws (Appendix B).

Contrast	Participants	Mean Δ	95% CI
<i>Later donor available: 1,001 participant-targets</i>			
LOSO - expanding window	138	+0.035	[+0.030, +0.040]
LOSO - session-count-matched reference	138	+0.025	[+0.022, +0.029]
Session-count-matched reference - expanding window	138	+0.009	[+0.005, +0.014]
<i>Strict-future common support: 560 participant-targets</i>			
LOSO - expanding window	138	+0.053	[+0.045, +0.061]
LOSO - session-count-matched reference	138	+0.040	[+0.035, +0.045]
Session-count-matched reference - expanding window	138	+0.013	[+0.006, +0.020]
Strict future - expanding window	138	+0.017	[+0.007, +0.026]

design at hand is more informative than a fixed correction applied to LOSO.

B. CONSEQUENCES FOR BENCHMARKING

MOABB's cross-session evaluation asks how well a decoder performs across the sessions of one participant [15], and LOSO answers that question. For a decoder updated from

the history available before each session, expanding-window validation gives the corresponding chronological estimate, and our paired results show how far apart the two estimates are. A benchmark addressing both questions can report both; the cross-validator we provide implements the chronological comparison within MOABB (Appendix A-F).

Target averaging also changes the reported contrast. In-

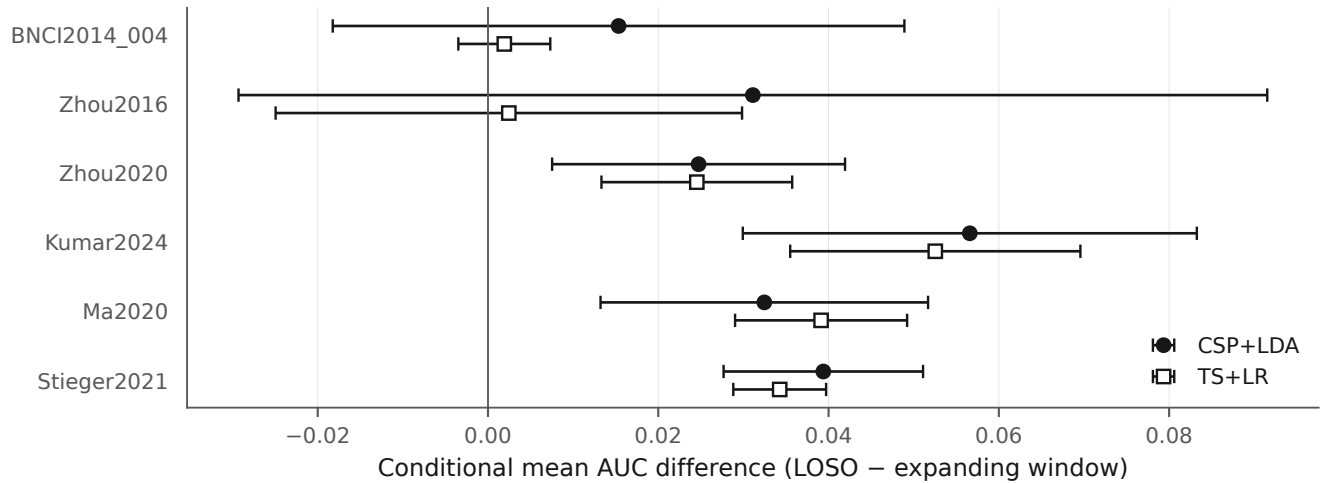


FIGURE 3. Conditional LOSO minus expanding-window AUC by dataset and decoder. Points are dataset means of participant-level contrasts, and bars are 95% confidence intervals across participants.

TABLE 4. Observed support by dataset: n is the number of participants; All origins, Cond. (conditional), and Strict future count participant-targets shared by the two decoders. The session-count-matched reference uses conditional support; the future-only arm requires strict-future support, $t \leq (T - 1)/2$. Mean donor-session count is the target-weighted expanding-window count t .

Dataset	n	Shared participant-targets		
		All origins	Cond.	Strict future
BNCI2014_004	9	36	27	18
Zhou2016	4	8	4	4
Zhou2020	20	119	99	59
Kumar2024	18	90	72	36
Ma2020	25	350	325	175
Stieger2021	62	536	474	268
Total	138	1139	1001	560
Mean donor-session count	—	5.36	4.96	2.93

cluding final sessions lowers the average because it adds structural zeros, and regrouping the Ma2020 blocks into days changes both the update event and the weight of the final target (Section V-D).

Although the protocols shifted performance levels, the dataset-level order of CSP+LDA and TS+LR did not change, and participant-level ranking changes occurred mainly near ties. Comparisons between classical and deep motor-imagery decoders have been summarized by headline accuracy differences [45], which do not show whether the ordering would hold under another protocol. A benchmark therefore needs separate summaries of how a protocol changes performance levels and how it changes decoder order. Rankings should come with their score margins, the participants they cover, and how the training runs were defined.

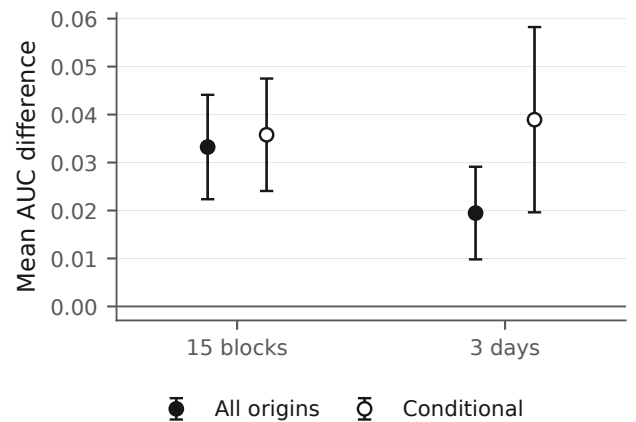


FIGURE 4. Ma2020 sensitivity to the temporal unit. The block analysis uses 15 ordered blocks; the day analysis concatenates five consecutive blocks within each of three recording days. Bars are 95% participant-level confidence intervals.

C. REPORTING PROCEDURE

A recent review notes that motor-imagery decoding studies often prioritize accuracy in isolation [46]. In BCI Competition IV data, including the cue-presentation and rest periods in the training data changed accuracy by more than ten percentage points for some participants, which complicates comparisons across studies [47]. To make such choices explicit, Table 5 lists five fields that pin down what a longitudinal estimate represents and fills them in for this analysis. LOSO answers the completed-trajectory question, and the expanding window answers the chronological one. When both matter, report both, and name each by what the decoder was allowed to see. In our materials, the result files identify participant, target, decoder, and protocol, and the support manifests say which rows and weights enter each estimate.

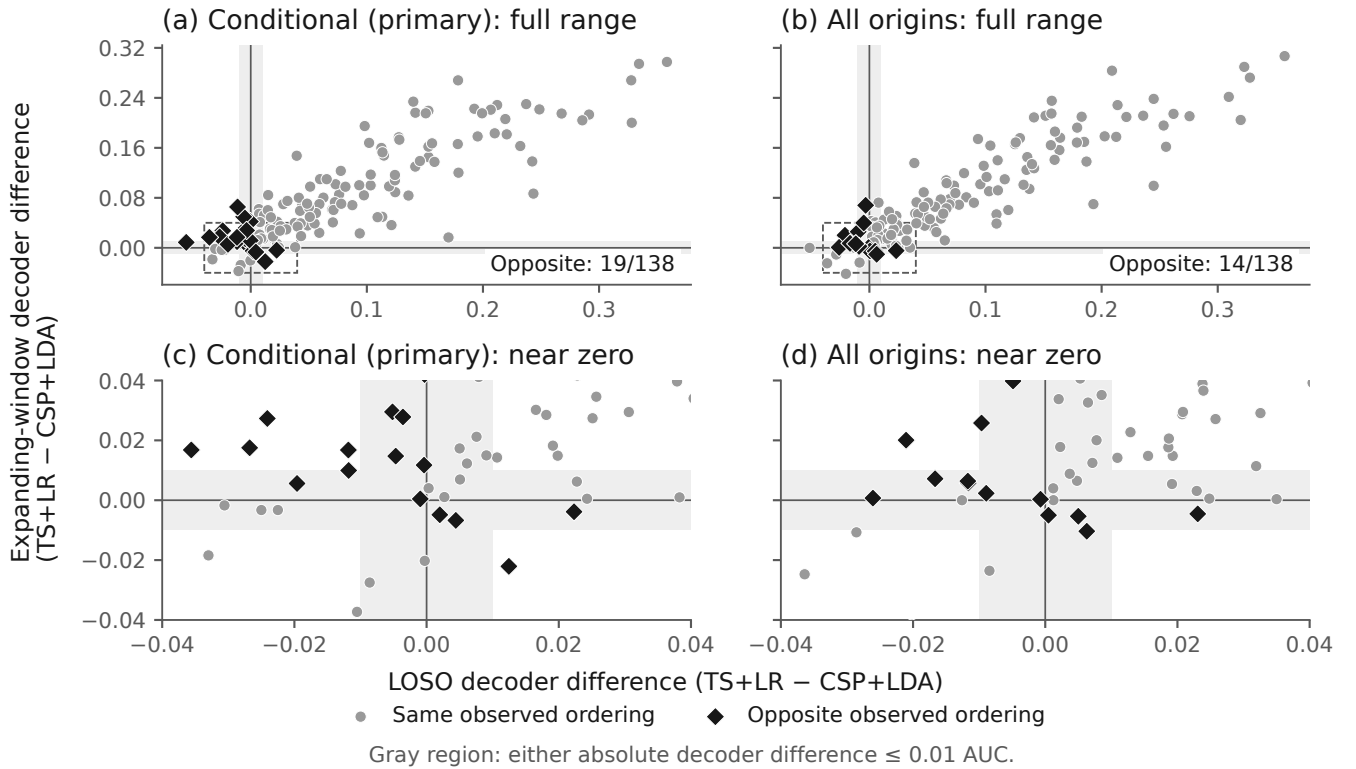


FIGURE 5. Participant-level decoder margins under the two protocols. Each point is one participant; positive margins favor TS+LR over CSP+LDA. Opposite-sign quadrants indicate a change in the observed higher-scoring decoder. The upper panels show the full range; the lower panels enlarge the dashed near-origin regions using identical limits. Gray circles indicate unchanged ordering and black diamonds indicate opposite ordering. Shaded bands mark absolute margins at or below 0.01 AUC.

TABLE 5. Reporting specification for longitudinal decoding evaluation, with this analysis as the worked example.

Reporting field	Specification in this audit
Prediction and update event	A fresh fit before each target session or block; the Ma2020 day-level analysis fits before each target recording day.
Available recordings and labels	Expanding window: all fully labeled prior sessions. LOSO: all fully labeled sessions other than the target.
Target contributions to fitting	No target contributions to learned parameters; fixed preprocessing followed by scoring on the target trials.
Temporal unit and endpoint	Source-defined session or block, through the final recorded unit; Ma2020 additionally regrouped by recording day.
Averaging population	Targets and two decoders averaged within participant, then participants weighted equally. The primary average uses targets with a later donor; the all-origin sensitivity summary also includes the final structural zero. Target-weighted and dataset-weighted summaries are identified separately.

VII. CONCLUSION

Across six motor-imagery datasets and 138 participants, LOSO scored 0.035 AUC above expanding-window validation on the same test targets, where later sessions were available. The difference was positive in every dataset but varied in size. When the LOSO pool was sampled down to the expanding window's number of sessions, most of the gap disappeared. The two decoders kept the same order in every dataset, and participant-level ranking changes occurred mainly near ties.

A protocol can therefore shift performance levels without changing which decoder ranks first, so the two need separate summaries. Benchmarks for decoders retrained before each session should report a chronological estimate that uses only

the history available at each update. They should also state the update event, temporal unit, and target averaging behind each number, and give score margins with every ranking.

DATA AVAILABILITY

This study is a secondary analysis of the datasets cited in Table 1. Access conditions and raw-data licenses are governed by the original repositories and dataset publications.

CODE AVAILABILITY

The analysis code, manuscript source, and result files will be released at <https://github.com/Yiming-S/loso-vs-prospective-bci>. The materials include `ForwardSessionSplitter`, a scikit-learn-compatible cross-validator accept-

ing per-trial session labels and an explicit session order. It implements expanding-window, immediately-previous, and first-session policies and can be passed to MOABB's `CrossSessionEvaluation` through the `cv_class` and `cv_kwargs` arguments, so the chronological protocol can be adopted without modifying the benchmark; Appendix A-F reports this check on BNCI2014_004. The materials also include version-pinned dependencies, the dataset registry, inclusion and support manifests, result-level CSV files, seeds, Stieger reuse provenance, figure sources, and LaTeX source. A result-only workflow regenerates the classical-decoder analyses and figures without raw EEG, and an audit checks result coverage, duplicate keys, score ranges, and final-origin structural zeros. Raw-cache checks, including epoch length and retained channels, require the original recordings or local caches.

ETHICS STATEMENT

The analysis uses previously released, de-identified datasets and collects no new human-participant data. Ethical approvals and consent procedures are reported in the original dataset publications.

AUTHOR CONTRIBUTIONS

Yiming Shen: conceptualization, methodology, software, validation, formal analysis, data curation, visualization, writing—original draft, and writing—review and editing.

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APPENDIX A ROBUSTNESS AND INTERPRETATION CHECKS

A. PARTICIPANT-LEVEL RANKING AND THRESHOLD SENSITIVITY

The primary participant-level median was +0.031 AUC (IQR [+0.017, +0.053]). Table 6 reports how many participants had a positive difference and how many exceeded each threshold. Table 7 reports observed ranking changes, margin sensitivity, and threshold crossings in both directions.

B. TARGET POSITION AND LATER-SESSION AVAILABILITY

Figure 6 plots the conditional difference against the number of later sessions that only LOSO could use. This count equals $T_i - 1 - t$, so each curve mixes target position, accumulated history, and trajectory length. The curves differ by dataset; in several of the longer trajectories the early targets show the larger differences.

TABLE 6. Observed participant-level magnitude and counts, n (%), of the protocol difference.

Metric	Conditional (primary)	All origins
Median	+0.031	+0.026
IQR	[+0.017, +0.053]	[+0.014, +0.046]
<i>Participants above each threshold, n (%)</i>		
$\Delta > 0$	124 (89.9%)	124 (89.9%)
$\Delta > 0.01$	115 (83.3%)	113 (81.9%)
$\Delta > 0.02$	98 (71.0%)	92 (66.7%)

TABLE 7. Observed participant-specific decoder ordering and threshold crossings ($n = 138$). Arrows go from LOSO to expanding-window ordering. Bilateral margins require both absolute TS+LR-minus-CSP+LDA mean AUC differences to exceed the threshold. Threshold crossings average both decoders and targets within participant.

Outcome	Conditional (primary)	All origins
Opposite ordering	19 (13.8%)	14 (10.1%)
CSP+LDA $\rightarrow$ TS+LR	15	10
TS+LR $\rightarrow$ CSP+LDA	4	4
<i>Opposite ordering with both margins above</i>		
0.005 AUC	11	7
0.01 AUC	6	1
0.02 AUC	1	1
<i>AUC ≥ 0.70 under one protocol only</i>		
LOSO only	16	13
Expanding window only	1	1
<i>AUC ≥ 0.75 under one protocol only</i>		
LOSO only	31	22
Expanding window only	0	0

TABLE 8. Alternative training policies and Ma2020 time-unit sensitivity.

Analysis	n	Mean Δ 95% CI
<i>Alternative policies (conditional targets): LOSO minus</i>		
Immediately previous session	138	+0.066 [+0.058, +0.074]
First session	138	+0.103 [+0.091, +0.116]
<i>Ma2020: blocks</i>		
All origins	25	+0.033 [+0.022, +0.044]
Conditional	25	+0.036 [+0.024, +0.048]
<i>Ma2020: days</i>		
All origins	25	+0.019 [+0.010, +0.029]
Conditional	25	+0.039 [+0.020, +0.058]

C. MA2020 TEMPORAL UNIT

Table 8 gives the Ma2020 block- and day-level estimates alongside two alternative training policies. On the day-level conditional support, the balanced-accuracy difference was +0.033 (95% CI [+0.019, +0.047]).

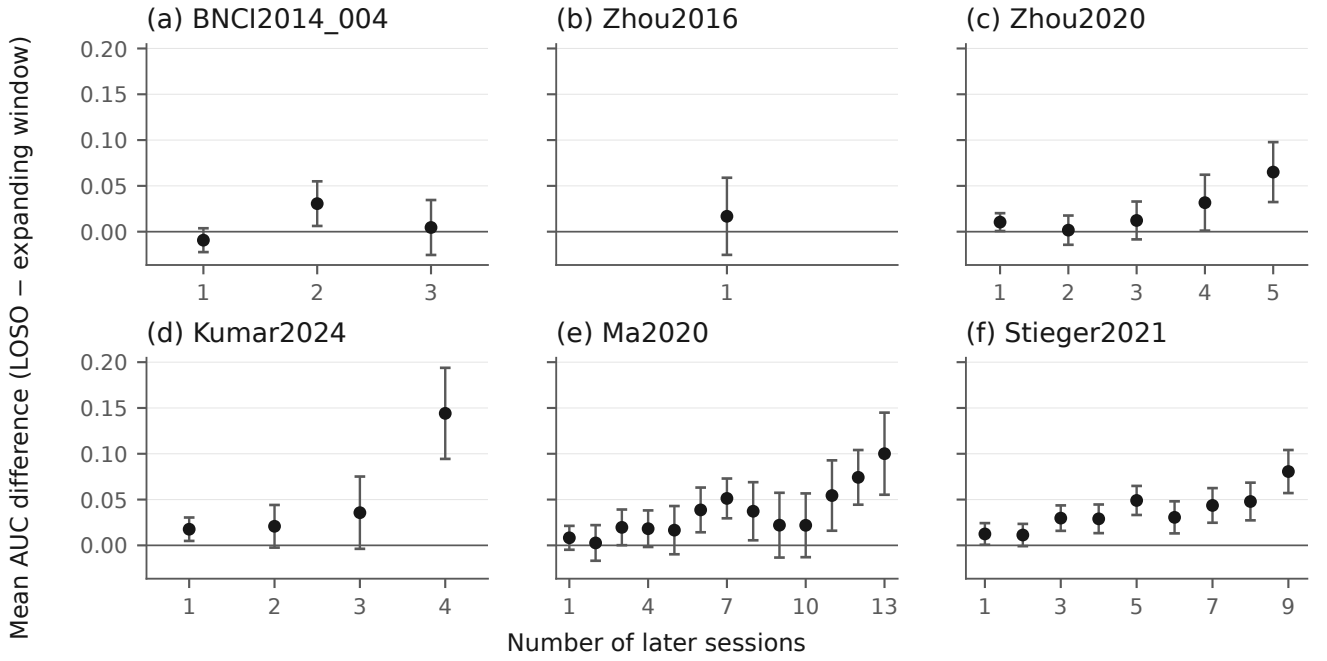


FIGURE 6. Dataset-specific association between the protocol difference and the number of later sessions available to LOSO. Each point averages participants and both decoders at a common within-dataset index; bars are 95% participant-level confidence intervals.

D. ADJACENT COVARIANCE-DISTANCE PROXY

Regressing the target-level contrast on the standardized distance between the unconditional mean covariances of adjacent sessions gave a participant-clustered coefficient of -0.0001 AUC (95% CI $[-0.0050, +0.0047]$, $p = 0.95$). Ninety-one of 2,002 target-decoder observations exceeded the Cook's-distance threshold; removing them produced a similarly near-zero association.

E. EEGNET CASE STUDY AND SEED SENSITIVITY ON STIEGER2021

We also analyzed an existing EEGNet run on Stieger2021 ($n = 62$, seed 20260716) that used the same filtering, resampling, epoching, and target definitions, with a seeded NumPy RandomState and PyTorch seed set before each fit. EEGNet [48] used the standard compact configuration ($F_1 = 8$, $D = 2$, $F_2 = 16$, temporal kernel 64, dropout 0.25), trained with Adam (learning rate 10^{-3} , weight decay 10^{-4} , batch size 32) for at most 250 epochs with early stopping (patience 40). Per-channel means and standard deviations were fitted on all donor trials. The donor trials were then randomly permuted, and $\max(2, \text{round}(0.15n_{\text{donor}}))$ of them were held out for validation with the rest used for optimization; if either subset had fewer than two classes, the complete donor pool served both. Early stopping monitored validation cross-entropy (minimum improvement 10^{-4}) and retained the best weights. No target trials contributed to standardization, fitting, or model selection.

Table 9 reports the result; paired comparisons use participant mean differences over common targets with two-sided t intervals and tests. Under LOSO, EEGNet scored

TABLE 9. Illustrative Stieger2021 analysis ($n = 62$, seed 20260716): participant-weighted mean AUC over common targets and the all-origin protocol difference. Bold entries identify the higher-scoring decoder.

Protocol	CSP+LDA	TS+LR	EEGNet
LOS0	0.730	0.861	0.882
Expanding window	0.696	0.831	0.801
LOS0 – expanding window	+0.034	+0.030	+0.081

0.021 AUC above TS+LR (95% CI $[0.006, 0.036]$, paired $p = 0.007$). Under the expanding window the order flipped: TS+LR scored 0.031 AUC above EEGNet (95% CI $[0.014, 0.047]$, $p < 0.001$). The protocol difference was largest for EEGNet, +0.081 AUC, roughly 2.5 times that of either classical decoder (+0.034 and +0.030). At the participant level the EEGNet-versus-TS+LR winner changed for 25 of 62 participants, and for 16 of them both within-protocol margins exceeded 0.01 AUC.

For the seed check we took a fixed random subset of 12 participants (seed 20260716), six with seven sessions and six with eleven, and retrained EEGNet with seeds 20260716, 20260717, and 20260718 on the same device and configuration. Each run was compared with TS+LR on the same participants and common targets ($t \geq 1$), averaging targets within participant and then weighting participants equally (Table 10). TS+LR scored higher under the expanding window at all three seeds, while the LOSO differences ranged from -0.0055 to $+0.0005$ AUC, inside the 0.01 near-tie band. At seed 20260716 the subset's scores match the full-cohort records.

TABLE 10. EEGNet-minus-TS+LR mean AUC on the same fixed Stieger2021 subset ($n = 12$), with participants weighted equally over common targets. Positive values favor EEGNet.

Training seed	LOSO	Expanding window
20260716	−0.0055	−0.0635
20260717	−0.0030	−0.0534
20260718	+0.0005	−0.0505

F. MOABB INTEGRATION CHECK

To check that the chronological protocol runs inside the unmodified benchmark, we handed the cross-validator to MOABB 1.5.0 by setting `cv_class` to `ForwardSessionSplitter` and `cv_kwargs` to `{'mode': 'expanding'}` in `CrossSessionEvaluation`, and ran BNCI2014_004 with the same paradigm settings (8–35 Hz, [0, 3] s, 128 Hz) and the two fixed decoders. The benchmark, with no change to its evaluation code, produced one score per target session $t \geq 1$: four expanding-window splits per participant, 36 scores per decoder, 72 in total. These agreed with the manuscript’s expanding-window AUCs for the same participant, target, and decoder to within 0.004 AUC (mean absolute difference below 0.001), and the benchmark’s default LOSO run agreed with the manuscript’s LOSO AUCs to the same degree. The small residual comes from epoch length: the benchmark keeps the sample at exactly 3 s (385 samples), whereas the audited loader crops to 384. Mean AUC over the 36 participant–targets was 0.784 for CSP+LDA and 0.808 for TS+LR in both implementations. The script and its per-session comparison table are part of the analysis materials.

APPENDIX B INTERVAL DEFINITIONS AND DONOR-SAMPLING UNCERTAINTY

A. DATASET-LEVEL SYNTHESIS

Let y_d and v_d be the participant mean and its estimated sampling variance for dataset d , and let $k = 6$. With restricted-maximum-likelihood estimate $\hat{\tau}^2$, the weights and pooled mean are

$$w_d = (v_d + \hat{\tau}^2)^{-1}, \quad \hat{\mu} = \frac{\sum_d w_d y_d}{\sum_d w_d}. \quad (7)$$

The reported mean interval uses the unscaled standard error and is $\hat{\mu} \pm t_{k-1, 0.975} (\sum_d w_d)^{-1/2}$. The approximate prediction interval [42] is $\hat{\mu} \pm t_{k-2, 0.975} \sqrt{\hat{\tau}^2 + (\sum_d w_d)^{-1}}$. For the Hartung–Knapp–Sidik–Jonkman sensitivity analysis [43], define $q = \sum_d w_d (y_d - \hat{\mu})^2 / (k - 1)$ and replace the mean standard error by $\sqrt{q / \sum_d w_d}$. This gives intervals of [+0.009, +0.039] for all origins and [+0.014, +0.047] for conditional targets.

B. DONOR-SAMPLING UNCERTAINTY OF THE MATCHED REFERENCE

The matched reference averages each row’s AUC over $B = 5$ random donor draws, so the participant scatter behind the

matched-reference intervals contains draw noise as well as participant variation; the unsampled LOSO–expanding contrast is untouched by the draws. Here we bound the draw part. The cached standard deviation $s_{r,0}$ of each row’s AUC across the draws uses divisor B , so the standard error of the row’s draw mean is $e_r = s_{r,0} / \sqrt{B - 1}$. The participant-weighted contrast is $\sum_r a_r \bar{x}_r$ with nonnegative weights a_r summing to one (equal participants, then equal targets and decoders within participant). Each participant draws from its own seed, so draw errors are independent across participants; only 3 of 135 participant–length pairs share a draw pattern across datasets. The intervals of Table 3 are computed from that scatter and are not adjusted for a shared draw component. Per-draw AUCs were not stored, so the covariance of draw errors across rows is not identifiable from the cached summaries. Under independent draws across targets, with the two decoder rows of one target treated as perfectly correlated, the donor-sampling component of the aggregate standard error is 0.0007 AUC on the conditional support and 0.0012 on the strict-future support, against a participant-level standard error of 0.0022 for the matched-minus-expanding contrast. The covariance-agnostic bound $\sum_r a_r e_r$ equals 0.0180 and 0.0221 AUC on the two supports.

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